

The Magic of MLOps Pipelines

This article explores MLOps pipelines that can be used in machine learning systems for model deployment. It also investigates why MLOps is necessary in today's machine learning system development, which demands faster model deployments, and how it helps to accomplish that.



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Gerd Altmann from pixabay.com

DeepLearning.ai reports that only 22 per cent of the companies using machine learning for solving their problems are actually able to deploy the model. In traditional software engineering projects, only the code keeps changing with new requirements and the addition of functionalities. On the other hand, in machine learning (ML) projects, data also keeps changing along with the code. Distribution of data also changes with this. For example, let's say we are building a machine learning model for predicting the total daily sales of an e-commerce platform. The distribution of data as

well as predictions for non-sale days will be very different from that of sale days. Thus, our model needs to be re-trained for the new data so that it can predict total sales on non-sale days as well. Thus, we see that as data and code change, the model too needs to be changed.

There are many other challenges that come with frequently changing data and models. These are as follows:

- We need to continuously monitor model performance and automate retraining of models when there is deterioration in performance.
- We need to develop processes to version models as well as data, and

keep records of how each version of the model performed with each version of data.

- We need a mechanism to quickly deploy new models, and to integrate the model changes with the code of the applications using it.
- MLOps helps us solve these problems with ease. MLOps translates to ML + DevOps. DevOps is a set of software engineering practices that allows us to deploy newer versions of software with agility. We need to deploy ML models with agility. That's where MLOps comes into the picture. It is a field that applies DevOps principles to ML systems.